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Permo-Triassic magnetostratigraphy in China: the type section near Taiyuan, Shanxi Province, North China

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SUMMARY

A palaeomagnetic study has been made of samples collected from the Permo-Triassic type section near Taiyuan, Shanxi Province in the North China Block. No results were obtained from the Upper Carboniferous Shanxi and Taiyuan Formations. Results from the Permian Lower and Upper Shihezi and Shiqianfeng Formations and from the Lower Triassic Liujiagou Formation were restricted mainly to the red shale and red mudstone horizons that were preferentially sampled for this study. After thermal demagnetization, the following pole positions were obtained: Lower Permian (Lower Shihezi Formation, $N = 11$ samples) 20.9°N, 1.4°E with $dp = 6.1°$ and $dm = 12.2°$; Upper Permian (combined Upper Shihezi and Shiqianfeng Formations, $N = 89$ samples) 47.8°N, 357.4°E with $dp = 2.3°$ and $dm = 4.3°$; Lower Triassic (Liujiagou Formation, $N = 25$ samples) 50.0°N, 351.0°E with $dp = 3.7°$ and $dm = 7.0°$. The sequence crosses the younger boundary of the Permo-Carboniferous reversed (Kiaman) superchron, and the polarity sequence observed suggests that this superchron ended in mid-Ufimian times. A minimum of 13 polarity zones (seven normal and six reverse) then follow before the Permo-Triassic boundary. This is in excellent agreement with currently available observations worldwide.

Key words: China, magnetostratigraphy, palaeomagnetism, Permian, Triassic.

INTRODUCTION

As part of a China–Australia cooperative programme in palaeomagnetism to study the Permo-Triassic type sections of the major blocks of China, we report here the details of a study of the type section at Ximing near Taiyuan, Shanxi Province (37.8°N, 112.3°E) in the North China Block (Fig. 1). The programme was carried out as a joint project between the palaeomagnetic laboratories of the Chinese Academy of Geological Sciences in Beijing, the Research School of Earth Sciences at the Australian National University (ANU) in Canberra and the CSIRO Division of Exploration Geoscience in Sydney. Initial results were reported by McElhinny *et al.* (1981) for the North and South China Blocks, and a brief report of the overall results has been given in Chinese by Ma & Zhang (1989). The final results from the South China Block have been reported by Ma *et al.* (1993). In this paper we report the details of the final results from the North China Block from our study of the entire succession at this locality. A study of Permo-Triassic sections in the northern part of the Tarim Block that also formed part of this China–Australia cooperative programme has been published in McFadden *et al.* (1988).

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GEOLOGY AND SAMPLING

A summary outline of the stratigraphy and sampling details of the Upper Carboniferous to Lower Triassic type section at Ximing near Taiyaun, Shanxi Province is given in Table 1. The section is situated across the south-east limb of the Xishan syncline and exposes a succession from the mid-Carboniferous Benxi Formation up to the Lower Triassic Liujiagou Formation. The Taiyuan and Shanxi Formations that overlie the Benxi Formation consist mainly of continental and marine white to grey sandstones with interbedded coal measures. The Shanxi Formation is rich in brachiopods, corals and fusilinids that date it as late Late Carboniferous. The Permian Shihezi

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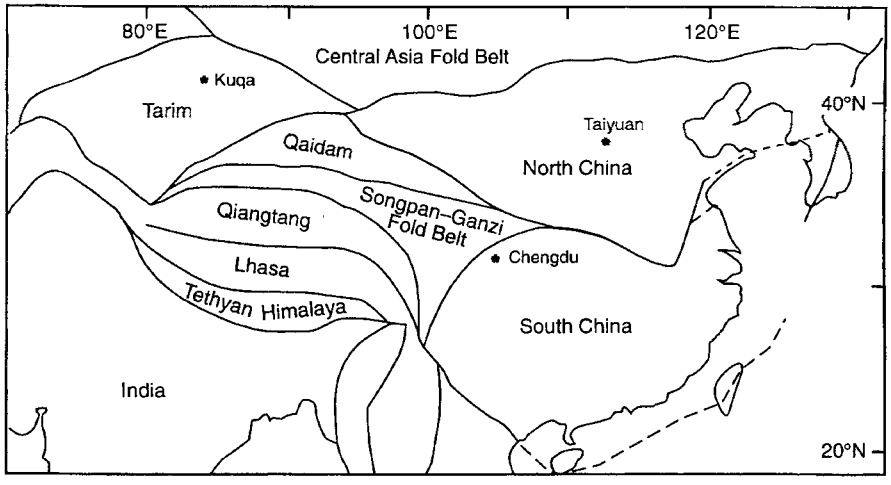


Figure 1. Simplified map showing the major blocks of Asia. The three Permo-Triassic successions sampled for the China–Australia cooperative programme are shown by asterisks. The North China Block succession was sampled near Taiyuan (this study), the South China Block succession was sampled near Emei Mountain (Ma *et al* 1993) and the Tarim Block succession was sampled near Kuqa (McFadden *et al.* 1988).

Table 1. Stratigraphic succession sampled at the Permo-Triassic type section near Taiyuan, Shanxi Province, North China. Layer (bed) numbers correspond to the official Chinese stratigraphic scheme for this section.

Formation		Stratigraphic Thickness (m)	Sampled Layers	No. of Samples
Lower Triassic	Liujiagou	400+	131	39
Upper Permian	Shiqianfeng	94	124-129	43
Upper Permian	Upper Shihezi C	203	104-117	83
Upper Permian	Upper Shihezi B	169	87-103	47
Upper Permian	Upper Shihezi A	72	71-84	16
Lower Permian	Lower Shihezi	114	61-68	24
Upper Carboniferous	Shanxi	33	45	6
Upper Carboniferous	Taiyuan	116	25-37	28

and Shiqianfeng Formations and Lower Triassic Liujiagou Formation conformably overlie these rocks. Whereas the Lower Permian sequence consists mainly of lighter coloured arkosic rocks, in the Upper Permian and Lower Triassic the sequence consists mainly of siltstones, shales and mudstones. The base of the Permian sequence includes some thin coal and bauxite beds, besides the argillaceous lithologies. In the upper parts of the section there are many relatively thin beds of red shales and mudstones that were sampled preferentially because these were likely to be the best material for palaeomagnetic measurements. The official Chinese stratigraphic scheme for this section divides it into a series of numbered layers (beds) of varying thicknesses, each of which corresponds to a successive change in rock type. Sampling commenced with layer 25 in the Upper Carboniferous Taiyuan Formation and ended at layer 131, which corresponds to the top of the succession, in the Lower Triassic Liujiagou Formation. 286 samples were collected from 33 selected layers, either as hand samples or as cores drilled with a petrol-driven portable sampling drill. Sampling details are given in Table 1. The section is tilted variably to the east and south-east with dips between 3° and 11° and has a stratigraphic thickness of 1300 m.

PALAEOMAGNETIC MEASUREMENTS

Laboratory measurements were carried out at the palaeomagnetic laboratory at the ANU in Canberra. Magnetic measurements were made using a Digico spinner magnetometer. All samples were subjected to stepwise thermal demagnetization up to 670 °C using the apparatus described by McElhinny, Luck & Edwards (1971). Data processing was carried out at the palaeomagnetic laboratory of the CSIRO in Sydney using orthogonal vector diagrams following Zijderveld (1967) and these were analysed using the principal component analysis method of Kirschvink (1980) and the linearity spectrum analysis method of Schmidt (1982).

All sandstone samples from the Upper Carboniferous Taiyuan and Shanxi Formations were either unstable or magnetized in the direction of the Earth's present field. This direction persisted up to 670 °C and therefore no useful palaeomagnetic information could be derived from these rocks. Initially, many of the Permian and Triassic samples also showed a magnetization close to the direction of the Earth's present field, but at temperatures above 400–500 °C a substantial Recent overprint was removed, revealing a univectorial characteristic component direction to the origin on a Zijderveld plot. Unfortunately many of the lighter coloured arkose samples did not reveal this characteristic component and the results presented here are thus restricted mainly to the red shale and mudstone samples collected from the succession. This may be the reason that Lin (1984) obtained only limited results from this same sequence, because his sampling did not include the redbeds that are predominant in our collection.

Some typical Zijderveld diagrams are illustrated in Fig. 2. The upper parts of the succession in the Shiqianfeng and Liujiagou Formations that cross the Permo-Triassic boundary are mainly normally magnetized, and Fig. 2(a) shows results from one sample with a univectorial component towards the origin downward dipping in a north-westerly direction. In Fig. 2(b) a north-west and downward-dipping direction, similar to that in Fig. 2(a), is observed as an overprint on a south-east and shallow-dipping component of nearly opposite polarity. This is a similar situation to that observed by Ma *et al.* (1993)

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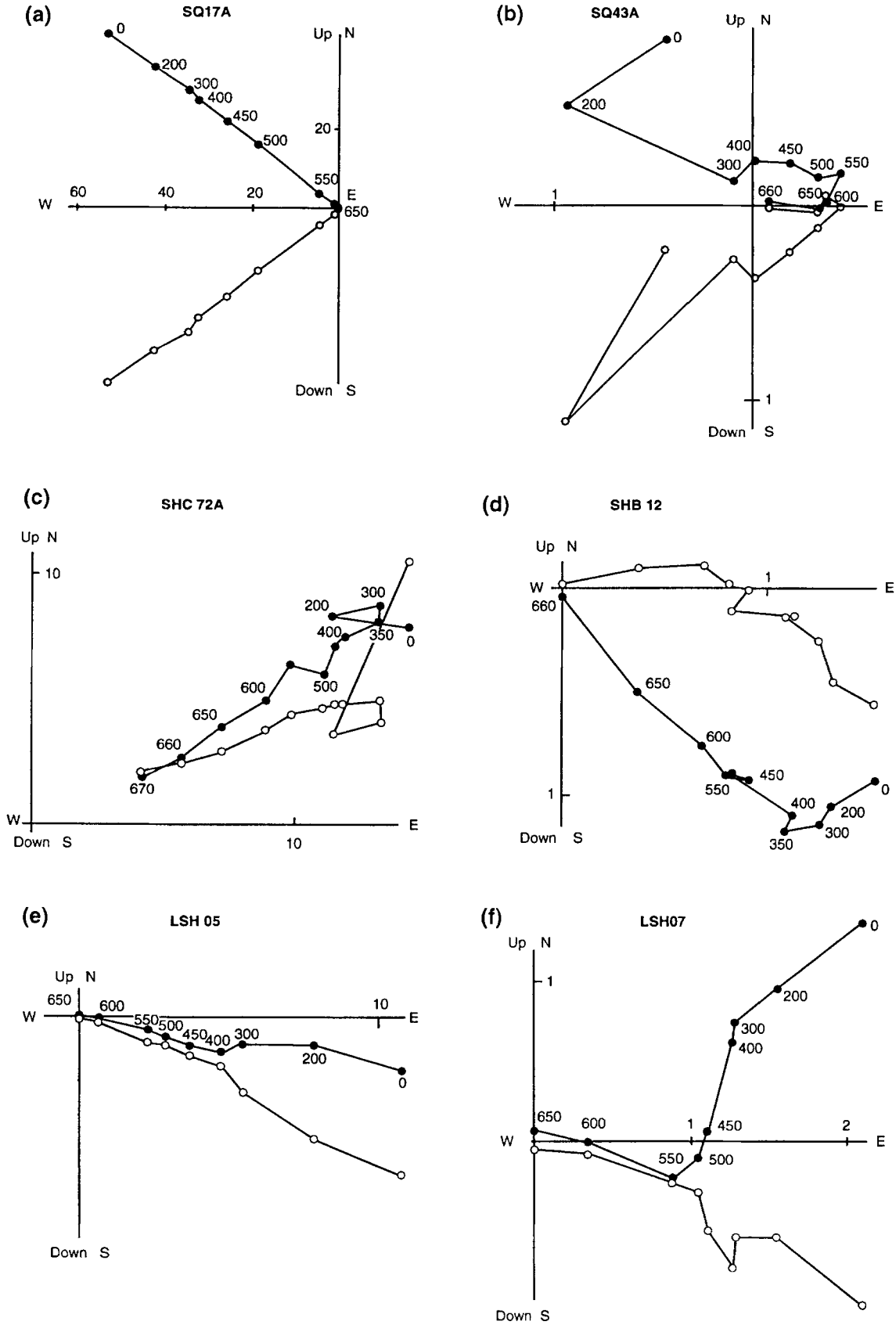


Figure 2. Examples of Zijderveld (1967) vector plots in stratigraphic coordinates for samples of the Lower and Upper Shihezi Formations during thermal demagnetization. Magnetizations are given in units of mA m^{-1}.

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Table 2. Formation mean directions of magnetization. N is the number of samples, D and I are the declination and inclination, respectively, of the mean direction of magnetization, k is the Fisher (1953) precision parameter and α_{95} is the radius of the circle of 95 per cent confidence about the mean direction. The preferred result for each grouping is given in bold type.

Formation	Position	Polarity	N	D	I	k	α_{95}
Liujiagou	**TRL**	**N**	**18**	**327.8**	**23.9**	**21.6**	**7.6**
	TRL	R	7	141.2	-19.6	17.3	14.9
	TRL	**N+R**	**25**	**325.9**	**22.8**	**20.4**	**6.6**
Shiqianfeng	PU	N	14	297.6	22.7	11.7	12.1
	PU	R	4	134.9	-25.0	20.8	20.6
	PU	N+R	18	301.5	23.4	12.4	10.2
Upper Shihezi	PU	N	17	332.5	28.0	22.9	7.6
	PU	R	54	143.8	-24.4	20.2	4.4
	PU	N+R	71	145.8	-25.4	20.2	3.8
Shiqianfeng and	PU	N	31	316.8	26.7	10.3	8.5
Upper Shihezi	PU	R	58	143.3	-24.5	20.7	4.2
Combined	**PU**	**N+R**	**89**	**141.2**	**-25.3**	**15.2**	**4.0**
Lower Shihezi	**PL**	**R**	**11**	**119.1**	**5.2**	**14.9**	**12.2**

in the Feixianguan Formation in Sichuan Province. It appears that during diagenesis a normally magnetized nearly opposite direction has been recorded as an overprint on a reversely magnetized component. In the Upper Shihezi Formation the characteristic reversely magnetized components are variably oriented to the south-east with shallow upward directions (Figs 2c and d), but the Lower Shihezi Formation is characterized by south-east and shallow downward directions (Figs 2e and f).

The overall results are summarized in Table 2, from which it can be seen that results were obtained only from about half of the samples collected and that these are almost entirely made up of the redbeds. The individual sample directions are shown on the stereonets in Fig. 3. In the Lower Triassic Liujiagou Formation, the mean normal and reverse components are 172.5° apart and they pass the reversals test of McFadden & McElhinny (1990) at level C. The underlying Shiqianfeng Formation (uppermost Permian) has normal- and reverse-component means that are 164.0° apart, but the reversals test is inconclusive. The Upper Shihezi Formation has normal and reverse mean directions that are 171.4° apart and pass the reversals test at level C. Because of the small number of samples with results from the Shiqianfeng Formation, we have combined the two Upper Permian Formations together as one result. In this case the mean normal and reverse components are now 173.7° apart and pass the reversals test at level B. This is our preferred result for the Upper Permian. Only 11 samples from the Lower Shihezi Formation gave any useful palaeomagnetic information so this result for the Lower Permian is not very well defined.

The sampling of this succession has been carried out preferentially from the red shale and mudstone horizons. We found no differences in results between the red and non-red horizons, and a comparison of our results in terms of pole positions with those from other localities in the North China

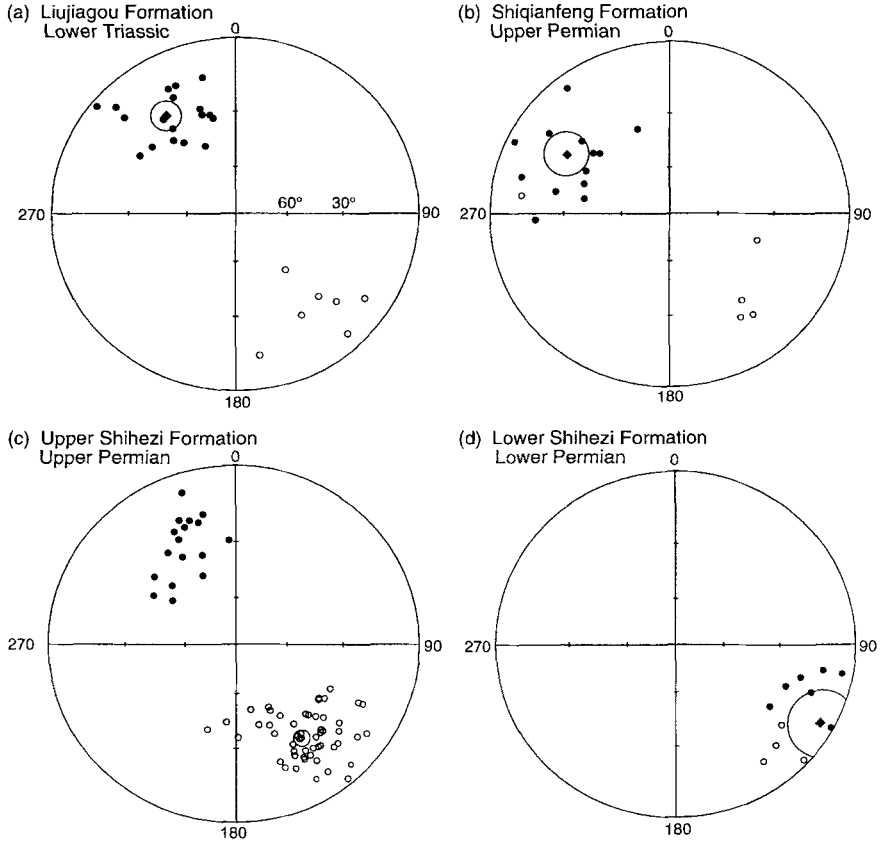


Figure 3. Sample mean directions observed in the various formations sampled, plotted in stratigraphic coordinates. (a) Liujiagou Formation, (b) Shiqianfeng Formation, (c) Upper Shihezi Formation, (d) Lower Shihezi Formation. The mean direction (diamond) and circle of 95 per cent confidence are shown for each formation. Equal-angle stereographic projection with solid (open) symbols plotted on the lower (upper) hemisphere.

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Block (see Table 4 below) shows that there is good consistency between rocks having the same stratigraphic position, regardless of rock type sampled, especially as most of the other results are not as restricted to redbed horizons as in this study. This provides strong evidence that the magnetization of the redbeds is primary.

PALAEOMAGNETIC POLE POSITIONS

Palaeomagnetic pole positions that correspond to the main results listed in Table 2 are given in Table 3. The Upper Permian and Lower Triassic poles are not significantly different, but there is a shift of more than 25° from the Lower Permian pole position, although this result is based on only 11 samples. In Table 4 we have summarized all the results to date from the Permo-Triassic of the North China Block and include only those results that were subjected to thermal demagnetization procedures. We have followed similar rejection criteria to those of Enkin *et al.* (1992) and have omitted the result by Lin (1984) from the Shiqianfeng Formation at Liaoning because the normal and reverse directions are not at all antipodal and it appears that there may be some effects resulting from local rotations. However, we are able to include some of the results of smaller collections (< 10 samples) because we then combine the data separately for the Lower Triassic and Upper Permian using the method for combining groups of palaeomagnetic directions or poles of McFadden & McElhinny (1995). In this method the result is essentially weighted towards those with the greatest number of measurements. In this case we have used sample numbers as the unit as in many of the studies the distinction between samples and sites when sampling a thick section (as we have done) is obscured. The Upper Permian mean is dominated by our result in this paper and the Lower Triassic mean is dominated by the excellent result reported by Yang *et al.* (1991).

The pole positions listed in Table 4 are plotted in Fig. 4, and, although each group of poles varies in latitude by 20° or so, they do show that there is a change of position between the Upper Permian and Lower Triassic of 12.2°, as shown by the difference between the mean poles for each group given in Table 4. This is comparable with the difference of 6.1° found by Enkin *et al.* (1992) between their calculated means. Note that the polar difference we find is a little less than the difference of 17.1° found between the same group of poles from

Table 3. Palaeomagnetic pole positions for the results given in Table 2. *N* is the number of samples and *dp* and *dm* are the semi-axes of the ovals of 95 per cent confidence about the pole position along the great circle from the site position and at right angles to the great circle, respectively. Preferred values are given in bold type.

Formation	Position	*N*	Lat. N	Long. E	*dp*	*dm*
Liujiagou	**TRL**	**25**	**50.0**	**351.0**	**3.7**	**7.0**
Shiqianfeng	PU	18	32.2	12.4	5.8	10.9
Upper Shihezi	PU	71	51.0	352.7	2.2	4.1
Shiqianfeng/Upper Shihezi Combined	**PU**	**89**	**47.8**	**357.4**	**2.3**	**4.3**
Lower Shihezi	**PL**	**11**	**20.9**	**1.4**	**6.1**	**12.2**

Table 4. Palaeomagnetic pole positions from studies of Permo-Triassic rocks in the North China Block in Shanxi and Shaanxi Provinces. Pol: polarity; *N*: number of samples. The mean North China pole positions are calculated using the method of combining groups of poles of McFadden & McElhinny (1995), where A_{95} is the radius of the circle of 95 per cent confidence about the calculated mean.

		Site Position				Pole Position		
Rock Unit	Position	Lat.N	Long.E	Pol	*N*	Lat.N	Long.E	Ref
Ordos Basin Seds	TRL	35.5	110.2	N+R	93	61.9	348.3	(2)
Liujiagou+Heshangou	TRL	38.6	112.0	N+R	35*	52.1	354.6	(3)
Liujiagou	TRL	39.1	110.0	N+R	15	55.7	355.3	(4)
Liujiagou	TRL	35.5	110.2	N	21	67.7	348.9	(5)
Liujiagou	TRL	37.8	112.3	N	7	45.4	6.8	(6)
Liujiagou	TRL	37.8	112.3	N+R	25	50.0	351.0	(1)
Mean North China	**TRL**	**A_{95} = 5.0°**		**N+R**	**196**	**58.2**	**351.6**	**(1)**
Upper Shihezi	PU	37.5	114.4	N+R	12	55.2	1.4	(7)
Upper Shihezi	PU	38.6	112.1	N+R	33	37.9	2.9	(7)
Upper Shihezi	PU	40.1	113.2	N+R	19	55.2	335.7	(7)
Upper Permian Seds	PU	39.3	98.8	N+R	43	41.0	358.6	(8)
Upper Shihezi	PU	39.1	110.0	R	9	48.6	357.0	(4)
U.Permian Volc+Seds	PU	42.8	118.9	N+R	62	51.9	341.7	(9)
Upper Shihezi	PU	35.5	109.1	R	20	33.2	18.5	(5)
Upper Shihezi	PU	37.8	112.3	R	10	38.1	6.3	(6)
Shiqianfeng + U.Shihezi	PU	37.8	112.3	N+R	89	47.8	357.4	(1)
Shiqianfeng + Liujiagou	PU-TRL	35.0	109.0	N+R	26	42.3	13.7	(10)
Mean North China	**PU**	**A_{95} = 6.2°**		**N+R**	**313**	**46.6**	**357.6**	**(1)**
Lower Shihezi	**PL**	**37.8**	**112.3**	**R**	**11**	**20.9**	**1.4**	**(1)**

* Number of samples estimated for seven sites. References: (1) This paper; (2) Yang *et al.* (1991); (3) Fang *et al.* (1988); (4) Yang (1987); (5) Wu *et al.* (1990); (6) Lin (1984); (7) Zhao & Coe (1989); (8) Meng, Coe & Frost (1989); (9) Zhao *et al.* (1990); (10) Cheng, Bai & Sun (1988).

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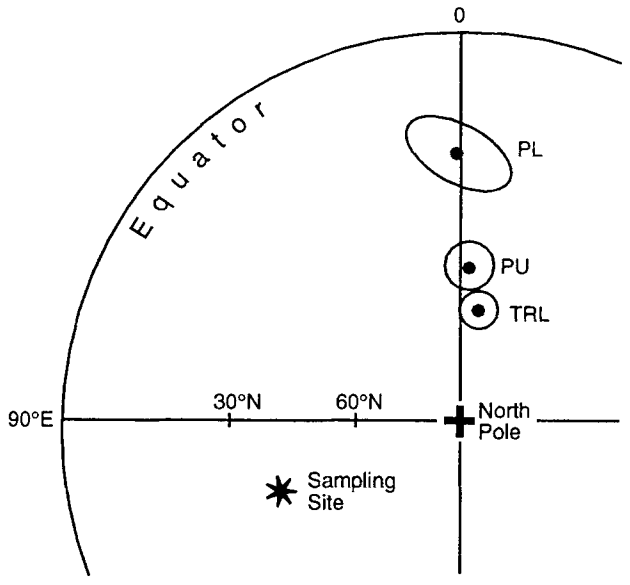


Figure 4. Mean Permian and Triassic pole positions for North China as calculated in Table 4 with circles (or oval) of 95 per cent confidence (PL: Lower Permian; PU: Upper Permian; TRL: Lower Triassic). Equal-angle polar stereographic projection.

the South China Block by Ma *et al.* (1993) or 19.9° found by Enkin *et al.* (1992). Our results for the Lower Permian, although based on only 11 samples, appear to be the only results for this age to date from the North China Block.

MAGNETOSTRATIGRAPHY

The results from this study are relevant for magnetostratigraphy. In Fig. 5 we give a stratigraphic interpretation of our results in terms of the sequence of reversals using the time-scale of Harland *et al.* (1990). The Upper Permian Shiqianfeng and Upper Shihezi Formations have not yet formally been assigned to the various stages of the Upper Permian by Chinese biostratigraphers. Therefore, because the sequence is continuous and no breaks in sedimentation are found, we have assigned the two formations to the stages shown in Fig. 5 on the basis of a uniform sedimentation rate through the Late Permain. On the time-scale of Harland *et al.* (1990) the Upper Permain only lasted for 11 myr, so that the margin for error in taking this approach is not very large. This approach essentially places the Shiqianfeng Formation in the Changxingian stage with the three members of the Upper Shihezi Formation divided into the Ufimian, Wordian, Capitanian and Longtanian stages.

The magnetostratigraphic column presented suggest that the end of the Permo-Carboniferous (Kiaman) superchron occurred in about mid-Ufimian times, corresponding to layer 73 of the sequence. Between the close of the Kiaman and the Permo-Triassic boundary 13 polarity zones (seven normal and six reverse) are observed. Because there are sampling gaps in the sequence, as indicated in Fig. 5, there could be more polarity changes present than are shown. Opdyke & Channell (1996) have recently summarized all available magnetostratigraphic information for this time period and their analysis indicates that there could be 14 polarity zones present. Their analysis also suggest that the end of the Kiaman superchron occurred at about mid-Ufimian times. All the various sections illustrated

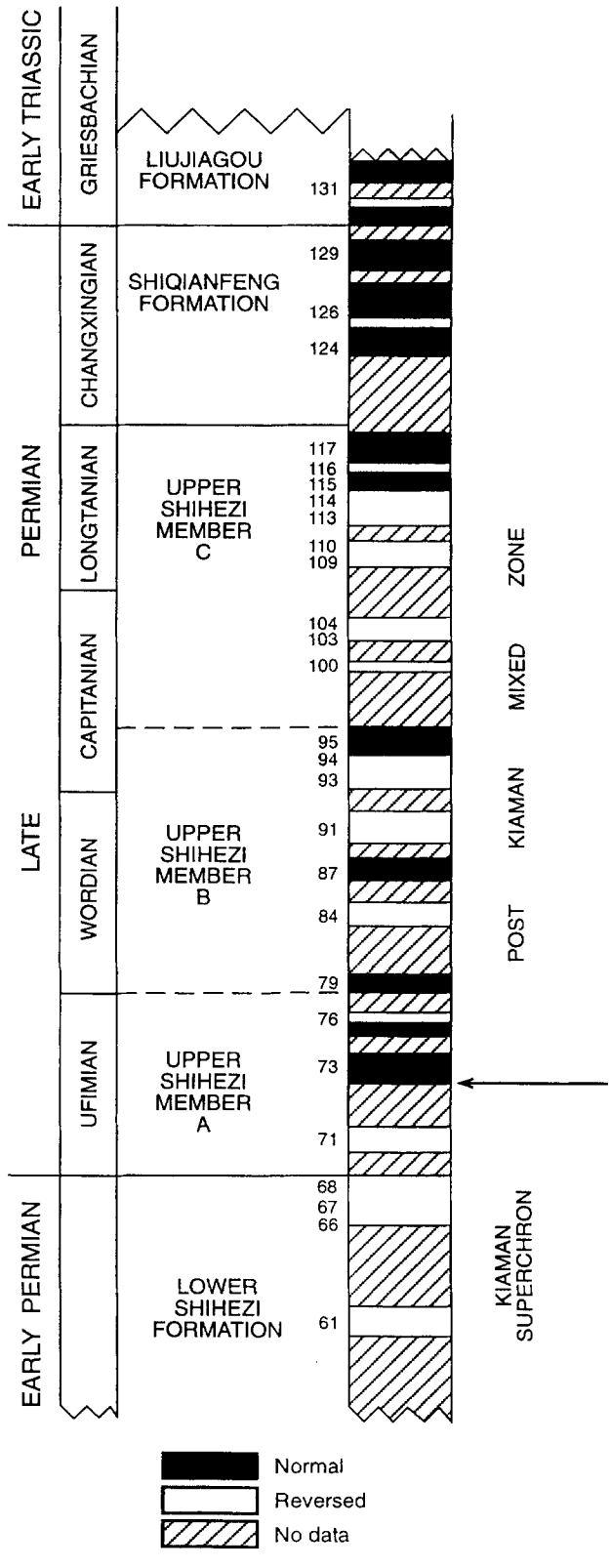


Figure 5. Polarity sequence observed in the Permo-Triassic type section near Taiyuan, Shanxi Province, North China. Normal/reverse polarity shown as black/white zones and the data gaps as diagonal lines. The Geologic Time Scale 1989 of Harland *et al.* (1990) is used.

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by Opdyke & Channell (1996) show the end of the Permian to be a zone of reverse polarity. This is not consistent with our observations, but then the precise positioning of the Permo-Triassic boundary between the Shiqianfeng and Liujiagou Formations could be open to question.

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